

Systemy plików

Studium przypadku

VBS - \$Boot

[unregistriert]		Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Drive D:	96% free	000000000	EB	52	90	4E	54	46	53	20	20	20	20	00	02	01	00	00	ër NTFS
File system:	NTFS	000000016	00	00	00	00	00	F8	00	00	3F	00	FF	00	20	00	00	00	ř...?..'...
[Read-only mode]		000000032	00	00	00	00	80	00	00	00	DF	5B	0F	00	00	00	00	00	€...ß[.....
		000000048	AD	1E	05	00	00	00	00	00	BF	AD	07	00	00	00	00	00	đ-.....
Alloc. of visible drive space:		000000064	02	00	00	00	00	00	00	00	E7	06	90	38	32	90	38	62	ç.ł828b
Cluster No.:	0	000000080	00	00	00	00	FA	33	C0	8E	D0	BC	00	7C	FB	B8	C0	07	ú3ŘŽĐŁ. ů,Ř.
\$Boot		000000096	8E	D8	E8	16	00	B8	00	0D	8E	C0	33	DB	C6	06	0E	00	ŽŘč...ŽŘ3ŮČ...
\		000000112	10	E8	53	00	68	00	0D	68	6A	02	CB	8A	16	24	00	B4	.čŠ.h...hj.ĚŠ.\$.'
Snapshot taken	21 min. ago	000000128	08	CD	13	73	05	B9	FF	FF	8A	F1	66	0F	B6	C6	40	66	.í.s.q`'Šňf.ŕČ0f
Physical sector No.:	32	000000144	0F	B6	D1	80	E2	3F	F7	E2	86	CD	C0	ED	06	41	66	0F	.ŕŮĚá?÷âťÍŘí.Af.
Logical sector No.:	0	000000160	B7	C9	66	F7	E1	66	A3	20	00	C3	B4	41	BB	AA	55	8A	·Ěf÷áfł.Ě'À»ŞŮŠ
		000000176	16	24	00	CD	13	72	0F	81	FB	55	AA	75	09	F6	C1	01	.\$.í.r.łŮŮŮ.óÁ.
Used space:	19,1 MB	000000192	74	04	FE	06	14	00	C3	66	60	1E	06	66	A1	10	00	66	t.č...Ěf`..f`..f
	20 001 280 bytes	000000208	03	06	1C	00	66	3B	06	20	00	0F	82	3A	00	1E	66	6A	f;..,..fj
Free space:	472 MB	000000224	00	66	50	06	53	66	68	10	00	01	00	80	3E	14	00	00	.fP.Sfh....€>...
	495 356 928 bytes	000000240	0F	85	0C	00	E8	B3	FF	80	3E	14	00	00	0F	84	61	00	čř'€>.....a.
Total capacity:	491 MB	000000256	B4	42	8A	16	24	00	16	1F	8B	F4	CD	13	66	58	5B	07	ˆBŠ.\$...<óí.fx[.
	515 358 208 bytes	000000272	66	58	66	58	1F	EB	2D	66	33	D2	66	0F	B7	0E	18	00	fXfX.ě-f3Ňf.....
		000000288	66	F7	F1	FE	C2	8A	CA	66	8B	D0	66	C1	EA	10	F7	36	f÷ňčĚŠĚf<ĐfÁę÷6
Bytes per cluster:	512	000000304	1A	00	86	D6	8A	16	24	00	8A	E8	C0	E4	06	0A	CC	B8	..+ŮŠ.\$.ŠčŘá..Ě.
Free clusters:	967 494	000000320	01	02	CD	13	0F	82	19	00	8C	C0	05	20	00	8E	C0	66	..í...ŠŘ. .ŽŘf
Total clusters:	1 006 559	000000336	FF	06	10	00	FF	0E	0E	00	0F	85	6F	FF	07	1F	66	61	'...o'..fa
Bytes per sector:	512	000000352	C3	A0	F8	01	E8	09	00	A0	FB	01	E8	03	00	FB	EB	FE	Ě ř.č.. ů.č..űęč
Sector count:	1 006 559	000000368	B4	01	8B	F0	AC	3C	00	74	09	B4	0E	BB	07	00	CD	10	'.<đ-<.t.'»...í.
Physical disk:	1	000000384	EB	F2	C3	0D	0A	42	88	A5	64	20	6F	64	63	7A	79	74	ěňĚ..BłĚd odczyt
		000000400	75	20	64	79	73	6B	75	00	0D	0A	42	72	61	6B	20	70	u dysku...Brak p
Mode:	Text	000000416	6C	69	6B	75	20	4E	54	4C	44	52	00	0D	0A	4E	54	4C	liku NTLDR...NTL
Character set:	ANSI ASCII	000000432	44	52	20	6A	65	73	74	20	73	6B	6F	6D	70	72	65	73	DR jest skompres
Offsets:	decimal	000000448	6F	77	61	6E	79	00	0D	0A	4E	61	63	69	98	6E	69	6A	owany...Naci[nij
Bytes per page:	34x16=544	000000464	20	43	74	72	6C	2B	41	6C	74	2B	44	65	6C	2C	20	61	Ctrl+Alt+Del, a
Window #:	1	000000480	62	79	20	7A	72	65	73	74	61	72	74	6F	77	61	86	0D	by zrestartowať.
No. of windows:	1	000000496	0A	00	00	00	00	00	00	00	83	98	AB	C6	00	00	55	AA	łł«Ć..UŞ

Byte Offset	Field Length	Field Name
0x00	3 bytes	Jump Instruction
0x03	8 bytes	OEM ID
0x0B	25 bytes	BPB
0x24	48 bytes	Extended BPB
0x54	426 bytes	Bootstrap Code
0x01FE	WORD	End of Sector Marker

Byte Offset	Field Length	Sample Value	Field Name
0x0B	WORD	0x0002	Bytes Per Sector
0x0D	BYTE	0x08	Sectors Per Cluster
0x0E	WORD	0x0000	Reserved Sectors
0x10	3 BYTES	0x000000	<i>always 0</i>
0x13	WORD	0x0000	<i>not used by NTFS</i>
0x15	BYTE	0xF8	Media Descriptor
0x16	WORD	0x0000	<i>always 0</i>
0x18	WORD	0x3F00	Sectors Per Track
0x1A	WORD	0xFF00	Number Of Heads
0x1C	DWORD	0x3F000000	Hidden Sectors
0x20	DWORD	0x00000000	<i>not used by NTFS</i>
0x24	DWORD	0x80008000	<i>not used by NTFS</i>
0x28	LONGLONG	0x4AF57F0000000000	Total Sectors
0x30	LONGLONG	0x0400000000000000	Logical Cluster Number for the file \$MFT
0x38	LONGLONG	0x54FF070000000000	Logical Cluster Number for the file \$MFTMirr
0x40	DWORD	0xF6000000	Clusters Per File Record Segment
0x44	DWORD	0x01000000	Clusters Per Index Block
0x48	LONGLONG	0x14A51B74C91B741C	Volume Serial Number
0x50	DWORD	0x00000000	Checksum

\$MFT

	[unregistriert]	Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Drive D:	96% free	171786240	46	49	4C	45	30	00	03	00	C5	28	30	00	00	00	00	00	FILE0...L(O.....
File system:	NTFS	171786256	01	00	01	00	38	00	01	00	98	01	00	00	00	04	00	00	8...l.....
[Read-only mode]		171786272	00	00	00	00	00	00	00	00	06	00	00	00	00	00	00	00	
		171786288	03	00	00	00	00	00	00	00	10	00	00	00	60	00	00	00	`...
Alloc. of visible drive space:		171786304	00	00	18	00	00	00	00	00	48	00	00	00	18	00	00	00	H.....
Cluster No.:	335520	171786320	E2	27	35	8C	B3	55	CA	01	E2	27	35	8C	B3	55	CA	01	à'5\$!UE.à'5\$!UE.
	\$MFT (#0)	171786336	E2	27	35	8C	B3	55	CA	01	E2	27	35	8C	B3	55	CA	01	à'5\$!UE.à'5\$!UE.
	\	171786352	06	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
Snapshot taken	6 hours ago	171786368	00	00	00	00	00	01	00	00	00	00	00	00	00	00	00	00	
Physical sector No.:	335552	171786384	00	00	00	00	00	00	00	00	30	00	00	00	68	00	00	00	O...h...
Logical sector No.:	335520	171786400	00	00	18	00	00	00	03	00	4A	00	00	00	18	00	01	00	J.....
		171786416	05	00	00	00	00	00	05	00	E2	27	35	8C	B3	55	CA	01	à'5\$!UE.
Used space:	19,1 MB	171786432	E2	27	35	8C	B3	55	CA	01	E2	27	35	8C	B3	55	CA	01	à'5\$!UE.à'5\$!UE.
	20 001 280 bytes	171786448	E2	27	35	8C	B3	55	CA	01	00	40	00	00	00	00	00	00	à'5\$!UE..@.....
Free space:	472 MB	171786464	00	40	00	00	00	00	00	00	06	00	00	00	00	00	00	00	.@.....
	495 356 928 bytes	171786480	04	03	24	00	4D	00	46	00	54	00	00	00	00	00	00	00	..\$.M.F.T.....
Total capacity:	491 MB	171786496	80	00	00	00	48	00	00	00	01	00	40	00	00	00	01	00	€...H.....@.....
	515 358 208 bytes	171786512	00	00	00	00	00	00	00	00	7F	00	00	00	00	00	00	00	l.....
		171786528	40	00	00	00	00	00	00	00	00	00	01	00	00	00	00	00	@.....
Bytes per cluster:	512	171786544	00	00	01	00	00	00	00	00	00	00	01	00	00	00	00	00	
Free clusters:	967 494	171786560	32	80	00	A0	1E	05	00	00	B0	00	00	00	48	00	00	00	2€.°...H...
Total clusters:	1 006 559	171786576	01	00	40	00	00	00	05	00	00	00	00	00	00	00	00	00	..@.....
Bytes per sector:	512	171786592	00	00	00	00	00	00	00	00	40	00	00	00	00	00	00	00	@.....
Sector count:	1 006 559	171786608	00	02	00	00	00	00	00	00	08	00	00	00	00	00	00	00	
Physical disk:	1	171786624	08	00	00	00	00	00	00	00	31	01	9F	1E	05	00	00	00	1.z.....
Mode:	hexadecimal	171786640	FF	FF	FF	FF	00	00	00	00	40	00	00	00	00	00	00	00	@.....
Character set:	ANSI ASCII	171786656	00	80	00	00	00	00	00	00	00	80	00	00	00	00	00	00	.€.€.
Offsets:	decimal	171786672	00	80	00	00	00	00	00	00	31	40	A0	1E	05	00	01	00	.€.1@
Bytes per page:	34x16=544	171786688	B0	00	00	00	48	00	00	00	01	00	40	00	00	00	05	00	°...H.....@.....
Window #:	1	171786704	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
No. of windows:	1	171786720	40	00	00	00	00	00	00	00	00	02	00	00	00	00	00	00	@.....
		171786736	08	00	00	00	00	00	00	00	08	00	00	00	00	00	03	00	

408

1024

6240+

408=

6640

6240+

1024=

7264

		Offset	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
Drive D:	[unregistriert]	171787248	00	00	00	00	00	00	00	00	00	00	00	00	00	00	03	00	
File system:	96% free NTFS	171787264	46	49	4C	45	30	00	03	00	98	0F	20	00	00	00	00	00	FILEO...].
[Read-only mode]		171787280	01	00	01	00	38	00	01	00	58	01	00	00	00	04	00	00	8...X.....
		171787296	00	00	00	00	00	00	00	00	04	00	00	00	01	00	00	00	
		171787312	03	00	00	00	00	00	00	00	10	00	00	00	60	00	00	00	`...
Alloc. of visible drive space:		171787328	00	00	18	00	00	00	00	00	48	00	00	00	18	00	00	00	H.....
Cluster No.:	335521	171787344	E2	27	35	8C	B3	55	CA	01	E2	27	35	8C	B3	55	CA	01	â'5Š±UE.â'5Š±UE.
	\$MFT (#0)	171787360	E2	27	35	8C	B3	55	CA	01	E2	27	35	8C	B3	55	CA	01	â'5Š±UE.â'5Š±UE.
	\	171787376	06	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
Snapshot taken	7 hours ago	171787392	00	00	00	00	00	01	00	00	00	00	00	00	00	00	00	00	
Physical sector No.:	335553	171787408	00	00	00	00	00	00	00	00	30	00	00	00	70	00	00	00	0...p...
Logical sector No.:	335521	171787424	00	00	18	00	00	00	02	00	52	00	00	00	18	00	01	00	R.....
Used space:	19.1 MB	171787440	05	00	00	00	00	00	05	00	E2	27	35	8C	B3	55	CA	01	â'5Š±UE.
	20 001 280 bytes	171787456	E2	27	35	8C	B3	55	CA	01	E2	27	35	8C	B3	55	CA	01	â'5Š±UE.â'5Š±UE.
Free space:	472 MB	171787472	E2	27	35	8C	B3	55	CA	01	00	10	00	00	00	00	00	00	â'5Š±UE.....
	495 356 928 bytes	171787488	00	10	00	00	00	00	00	00	06	00	00	00	00	00	00	00	
Total capacity:	491 MB	171787504	08	03	24	00	4D	00	46	00	54	00	4D	00	69	00	72	00	..\$.M.F.T.M.i.r.
	515 358 208 bytes	171787520	72	00	00	00	00	00	00	00	80	00	00	00	48	00	00	00	r.....€...H...
Bytes per cluster:	512	171787536	01	00	40	00	00	00	01	00	00	00	00	00	00	00	00	00	..@.....
Free clusters:	967 494	171787552	07	00	00	00	00	00	00	00	40	00	00	00	00	00	00	00	@.....
Total clusters:	1 006 559	171787568	00	10	00	00	00	00	00	00	00	10	00	00	00	00	00	00	
Bytes per sector:	512	171787584	00	10	00	00	00	00	00	00	31	08	EF	AD	07	00	00	00	1.d-....
Sector count:	1 006 559	171787600	FF	FF	FF	FF	00	00	00	00	15	00	00	00	3F	AD	14	62	?-.b
Physical disk:	1	171787616	DA	16	EB	23	82	8B	A6	28	EB	03	00	00	01	02	00	00	Ú.ë#,< (ë.....
		171787632	00	00	00	05	20	00	00	00	20	02	00	00	00	00	00	00	
Mode:	hexadecimal	171787648	80	00	00	00	48	00	00	00	01	00	40	00	00	00	01	00	€...H.....@.....
Character set:	ANSI ASCII	171787664	00	00	00	00	00	00	00	00	07	00	00	00	00	00	00	00	
Offsets:	decimal	171787680	40	00	00	00	00	00	00	00	00	10	00	00	00	00	00	00	@.....
Bytes per page:	34x16=544	171787696	00	10	00	00	00	00	00	00	00	10	00	00	00	00	00	00	
Window #:	1	171787712	31	08	EF	AD	07	00	00	00	FF	FF	FF	FF	00	00	00	00	1.d-....
No. of windows:	1	171787728	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
		171787744	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	
		171787760	00	00	00	00	00	00	00	00	00	00	00	00	00	00	03	00	

NTFS

Nadrzędna tablica plików

Plik 0	MFT
Plik 1	Fragment MFT (kopia)
Plik 2	Pliki systemowe (NTFS metadata files)
Plik 16	Pliki i katalogi użytkownika

Rekord plikowy

nagłówek
atrybut
atrybut
atrybut
znacznik końca rekordu plikowego:FFFFFFFFH

Atrybut standardowy

171814960	05 00 00 00 00 00 00 00	10 00 00 00 60 00 00 00	`...
171814976	00 00 00 00 00 00 00 00	48 00 00 00 18 00 00 00	H.....
5576 171814992	44 00 18 26 1F 56 CA 01	64 43 6A 2A 1F 56 CA 01	D..&.VE_.dCj*.VE_.
#28) 171815008	64 43 6A 2A 1F 56 CA 01	64 43 6A 2A 1F 56 CA 01	dCj*.VE_.dCj*.VE_.
ik.txt 171815024	20 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00	
: ago 171815040	00 00 00 00 02 01 00 00	00 00 00 00 00 00 00 00	
5608 171815056	00 00 00 00 00 00 00 00	30 00 00 00 70 00 00 00	0...p...

Nagłówek atrybutu – 24 bajty

Po nagłówku daty: czas utworzenia (0), czas ostatniej modyfikacji danych(8),
czas modyfikacji MFT (10h), czas ostatniego dostępu (18h)

Atrybut nazwy pliku

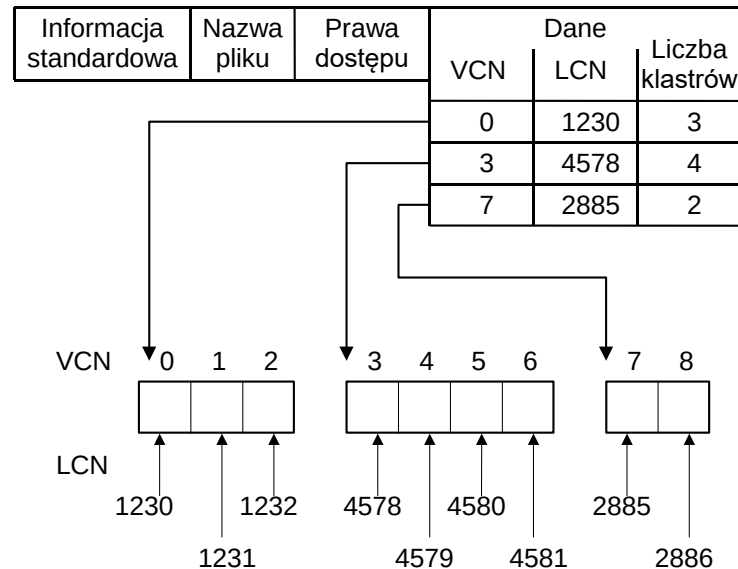
8	171815056	00 00 00 00 00 00 00 00	30 00 00 00 70 00 00 00	O...p...
6	171815072	00 00 00 00 00 00 02 00	52 00 00 00 18 00 01 00	R.....
	171815088	05 00 00 00 00 00 05 00	44 00 18 26 1F 56 CA 01	D...&.VE.
B	171815104	44 00 18 26 1F 56 CA 01	44 00 18 26 1F 56 CA 01	D...&.VE.D...&.VE.
s	171815120	44 00 18 26 1F 56 CA 01	00 00 00 00 00 00 00 00	D...&.VE.....
B	171815136	00 00 00 00 00 00 00 00	20 00 00 00 00 00 00 00	
s	171815152	08 03 70 00 6C 00 69 00	6B 00 2E 00 74 00 78 00	..p.l.i.k...t.x.
B	171815168	74 00 00 00 00 00 00 00	80 00 00 00 30 00 00 00	t.....€...O...

- nazwę pliku (offset 42H), cztery pola określające czas, faktyczny i zaalokowany rozmiar pliku (pola o offsetach 30H i 28H), znaczniki (offset 38H: tylko do odczytu, ukryty, systemowy, skompresowany, zaszyfrowany, itp.) nazwa pliku kodowana jest w standardzie Unicode, przy czym ilość znaków określa 1-bajtowe pole o offsecie 40H;

Atrybut DATA

MB	171815168	74 00 00 00 00 00 00 00	80 00 00 00 30 00 00 00	t.....€...0...
tes	171815184	00 00 18 00 00 00 03 00	15 00 00 00 18 00 00 00	
012	171815200	54 6F 20 6A 65 73 74 20	70 6C 69 6B 20 74 65 6B	To jest plik tek
094	171815216	73 74 6F 77 79 00 00 00	FF FF FF FF 82 79 47 11	stowy...'',yG.

w przypadku małych plików (o rozmiarze do 1 kiB) może to być zawartość pliku;
w przypadku większych plików podane są wskaźniki na dane



Modyfikacja pliku PE

- Wyszukanie pliku (FindFirstFile, FindNextFile)
- Otwarcie pliku (CreateFile, ReadFile/WriteFile, CreateFileMapping oraz MapViewOfFile)
- Weryfikacja formatu pliku (MZ, PE, modyfikacja)
- Wyznaczenie danych ostatniej sekcji
- Modyfikacja danych w nagłówku sesji
- Modyfikacje danych w nagłówku PE
- Doczepienie kodu
- Uzupełnienie sekcji
- Zapis pliku

Odzworowywanie plików

```
HANDLE    hPlik, hMap;
char * p;
DWORD rozmiar;
hPlik = CreateFile ("d:\\Builder\\teksty1.txt",
                    GENERIC_READ | GENERIC_WRITE,
                    0, NULL, OPEN_ALWAYS,
                    FILE_ATTRIBUTE_NORMAL, 0);

rozmiar = GetFileSize (hPlik, NULL);
hMap = CreateFileMapping(hPlik, NULL, PAGE_READONLY, 0, 0, NULL);
p = (char *) MapViewOfFile (hMap, FILE_MAP_ALL_ACCESS, 0, 0, 0);
*(p+10) = '#';
    *(p+40) = '#';
    *(p+rozmiar-1) = '\\0';
FlushViewOfFile(p, dwViewSize);
UnmapViewOfFile(p);
CloseHandle(hMap);
CloseHandle (hPlik);
```

- Sprawdzamy dwa pierwsze bajty, aby stwierdzić, czy mamy do czynienia z plikiem EXE MZ
- Jeśli tak, pobieramy adres PE header i ustawiamy bazę na początek nagłówka
- Sprawdzamy pierwsze dwa bajty, aby stwierdzić, czy rzeczywiście mamy do czynienia z nagłówkiem PE (linie 181 i 182).
- Jeśli tak, sprawdzamy wartość pola Checksum (offset 58H, DWORD)
- Jeśli plik nie był modyfikowany – przystępujemy do kroku następnego

- Rozmiar PE header jest stały i wynosi 18h.
- Zaraz za nagłówkiem PE występuje Optional header, którego rozmiar mamy pod offsetem 14h względem początku PE header.
- Suma wskazuje na początek Section header.
- Odczytać liczbę sekcji i przeszukiwanie w celu znalezienia tej, która ma zapisaną największą wartość w polu PointerToRawData (offset 14h względem Section header – czyli jest ostatnia w pliku) .
- Każdy opis sekcji ma 28H bajtów
- Początek Section header + numer sekcji razy 28h daje opis sekcji, którą należy zmodyfikować. (W przykładzie ESI -> PE, EDI -> nagłówek sekcji)

Uwaga: wpis dotyczący ostatniej sekcji pliku PE nie musi być ostatnim wpisem w Section header.